

Module - 4

Circuit Analysis in the S domain

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Introduction to Laplace Transform

- The circuits are characterised by the differential equations.
- The differential equation is turned into algebraic equations in Laplace transforms.
- It transforms the circuit from the time domain to frequency domain, obtain the solution and apply the inverse Laplace transform to the result to transform it back to the time domain.
- **Merits:**
 - It can be applied to a wider variety of inputs than phasor analysis.
 - It provides an easy way to solve circuit problems involving initial conditions
 - It provides solution in single operation.

Definition of the Laplace Transform

The Laplace transform is an integral transformation of a function $f(t)$ from the time domain into the complex frequency domain, giving $F(s)$.

Given a function $f(t)$, its Laplace transform, denoted by $F(s)$ or $\mathcal{L}[f(t)]$, is defined by

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt$$

where s is a complex variable given by

$$s = \sigma + j\omega$$

Inverse Laplace Transform

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j} \int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} F(s)e^{st} ds$$

Laplace transform pair

$$f(t) \quad \Leftrightarrow \quad F(s)$$

Laplace Transform Pairs

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s + a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
te^{-at}	$\frac{1}{(s + a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s + a)^{n+1}}$

$f(t)$	$F(s)$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

Properties of the Laplace Transform

Property	$f(t)$	$F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1 F_1(s) + a_2 F_2(s)$
Scaling	$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Time shift	$f(t - a)u(t - a)$	$e^{-as} F(s)$
Frequency shift	$e^{-at} f(t)$	$F(s + a)$
Time differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
	$\frac{d^2 f}{dt^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
	$\frac{d^3 f}{dt^3}$	$s^3 F(s) - s^2 f(0^-) - sf'(0^-) - f''(0^-)$
	$\frac{d^n f}{dt^n}$	$s^n F(s) - s^{n-1} f(0^-) - s^{n-2} f'(0^-) - \dots - f^{(n-1)}(0^-)$

Property	$f(t)$	$F(s)$
Time integration	$\int_0^t f(x) dx$	$\frac{1}{s} F(s)$
Frequency differentiation	$tf(t)$	$-\frac{d}{ds} F(s)$
Frequency integration	$\frac{f(t)}{t}$	$\int_s^\infty F(s) ds$
Time periodicity	$f(t) = f(t + nT)$	$\frac{F_1(s)}{1 - e^{-sT}}$
Initial value	$f(0)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s \rightarrow 0} sF(s)$
Convolution	$f_1(t) * f_2(t)$	$F_1(s)F_2(s)$

Numerical - 1

Determine the Laplace transform of each of the following functions:

(a) $u(t)$, (b) $e^{-at}u(t)$, $a \geq 0$, and (c) $\delta(t)$.

Solution:

(a) For the unit step function $u(t)$,

$$\begin{aligned}\mathcal{L}[u(t)] &= \int_{0^-}^{\infty} 1e^{-st} dt = -\frac{1}{s}e^{-st} \Big|_0^{\infty} \\ &= -\frac{1}{s}(0) + \frac{1}{s}(1) = \frac{1}{s}\end{aligned}$$

(b) For the exponential function,

$$\begin{aligned}\mathcal{L}[e^{-at}u(t)] &= \int_{0^-}^{\infty} e^{-at} e^{-st} dt \\ &= -\frac{1}{s+a} e^{-(s+a)t} \bigg|_0^{\infty} = \frac{1}{s+a}\end{aligned}$$

(c) For the unit impulse function,

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t) e^{-st} dt = e^{-0} = 1$$

Numerical - 2

Obtain the Laplace transform of $f(t) = \delta(t) + 2u(t) - 3e^{-2t}u(t)$.

Solution:

By the linearity property,

$$\begin{aligned} F(s) &= \mathcal{L}[\delta(t)] + 2\mathcal{L}[u(t)] - 3\mathcal{L}[e^{-2t}u(t)] \\ &= 1 + 2\frac{1}{s} - 3\frac{1}{s+2} = \frac{s^2 + s + 4}{s(s+2)} \end{aligned}$$

Numerical - 3

Determine the Laplace transform of $f(t) = t^2 \sin 2t u(t)$.

Solution:

$$\mathcal{L}[\sin 2t] = \frac{2}{s^2 + 2^2}$$

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n F(s)}{ds^n}$$

Using frequency differentiation

$$F(s) = \mathcal{L}[t^2 \sin 2t] = (-1)^2 \frac{d^2}{ds^2} \left(\frac{2}{s^2 + 4} \right)$$

$$\left(\frac{u}{v} \right)' = \frac{u'v - uv'}{v^2}$$

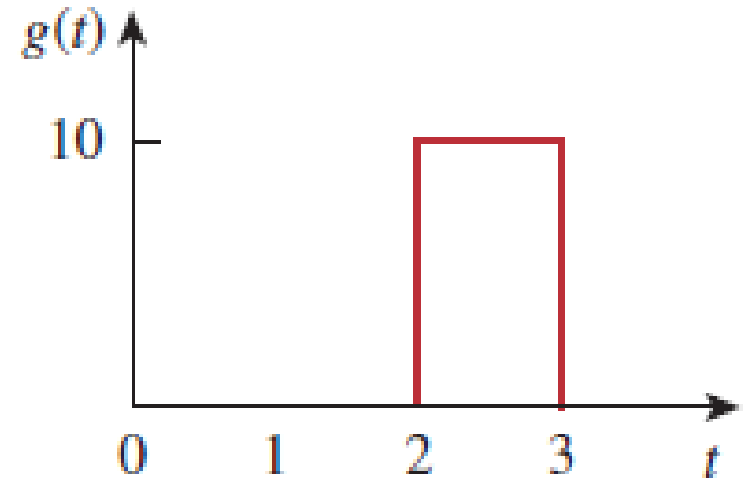
$$= \frac{d}{ds} \left(\frac{-4s}{(s^2 + 4)^2} \right) = \frac{12s^2 - 16}{(s^2 + 4)^3}$$

Numerical - 4

Find the Laplace transform of the gate function

Solution:

$$g(t) = 10[u(t - 2) - u(t - 3)]$$



Since we know the Laplace transform of $u(t)$, we apply the time-shift property and obtain

$$G(s) = 10\left(\frac{e^{-2s}}{s} - \frac{e^{-3s}}{s}\right) = \frac{10}{s}(e^{-2s} - e^{-3s})$$

The Inverse Laplace Transform

- If $F(s)$ has the general form of

$$F(s) = \frac{N(s)}{D(s)}$$

- Where $N(s)$ is the numerator polynomial and $D(s)$ is the denominator polynomial.
- The roots of $N(s) = 0$ are called the zeros of $F(s)$
- The roots of $D(s) = 0$ are called as the poles of $F(s)$.

Steps to find the Inverse Laplace Transform:

- Decompose $F(s)$ into simple terms using partial fraction expansion
- Find the inverse of each term by matching entries in Laplace transform pairs table.

Simple Poles

- If $F(s)$ has only simple poles (first order poles), then $D(s)$ becomes a product of factors

$$F(s) = \frac{N(s)}{(s + p_1)(s + p_2) \cdots (s + p_n)}$$

- Where $s = -p_1, -p_2, \dots, -p_n$ are the simple poles, and p_i is not equal to p_j , the poles are distinct. $p_i \neq p_j$
- Assuming the degree of $N(s)$ is less than the degree of $D(s)$, we use partial fraction expansion to decompose $F(s)$.

$$F(s) = \frac{k_1}{s + p_1} + \frac{k_2}{s + p_2} + \cdots + \frac{k_n}{s + p_n}$$

- The expansion coefficients $k_1, k_2, \dots, k_n$ are known as residues of $F(s)$.
- To find the expansion coefficients, we use residue method (Heaviside's theorem).

$$k_i = (s + p_i)F(s) \big|_{s=-p_i}$$

- Once the values of k_i are known, apply inverse of $F(s)$.

$$f(t) = (k_1 e^{-p_1 t} + k_2 e^{-p_2 t} + \cdots + k_n e^{-p_n t})u(t)$$

Numerical - 5

Find the inverse Laplace transform of

$$F(s) = \frac{3}{s} - \frac{5}{s+1} + \frac{6}{s^2+4}$$

Solution:

The inverse transform is given by

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}[F(s)] \\ &= \mathcal{L}^{-1}\left(\frac{3}{s}\right) - \mathcal{L}^{-1}\left(\frac{5}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{6}{s^2+4}\right) \\ &= (3 - 5e^{-t} + 3 \sin 2t)u(t), \quad t \geq 0 \end{aligned}$$

Numerical - 6

Find $f(t)$ given that

$$F(s) = \frac{s^2 + 12}{s(s + 2)(s + 3)}$$

Solution:

$$\frac{s^2 + 12}{s(s + 2)(s + 3)} = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s + 3}$$

where A , B , and C are the constants to be determined.

■ **METHOD 1** Residue method:

$$A = sF(s) \big|_{s=0} = \frac{s^2 + 12}{(s + 2)(s + 3)} \bigg|_{s=0} = \frac{12}{(2)(3)} = 2$$

$$B = (s + 2)F(s) \big|_{s=-2} = \frac{s^2 + 12}{s(s + 3)} \bigg|_{s=-2} = \frac{4 + 12}{(-2)(1)} = -8$$

$$C = (s + 3)F(s) \big|_{s=-3} = \frac{s^2 + 12}{s(s + 2)} \bigg|_{s=-3} = \frac{9 + 12}{(-3)(-1)} = 7$$

$$F(s) = \frac{2}{s} - \frac{8}{s + 2} + \frac{7}{s + 3}$$

By finding the inverse transform of each term, we obtain

$$f(t) = (2 - 8e^{-2t} + 7e^{-3t})u(t)$$

Repeated Poles

Suppose $F(s)$ has n repeated poles at $s = -p$. Then we may represent $F(s)$ as

$$F(s) = \frac{k_n}{(s + p)^n} + \frac{k_{n-1}}{(s + p)^{n-1}} + \cdots + \frac{k_2}{(s + p)^2} \\ + \frac{k_1}{s + p} + F_1(s)$$

where $F_1(s)$ is the remaining part of $F(s)$ that does not have a pole at $s = -p$. We determine the expansion coefficient k_n as

$$k_n = (s + p)^n F(s) \big|_{s=-p}$$

$$k_{n-1} = \frac{d}{ds}[(s + p)^n F(s)] \big|_{s=-p}$$

Repeating this gives

$$k_{n-2} = \frac{1}{2!} \frac{d^2}{ds^2}[(s + p)^n F(s)] \big|_{s=-p}$$

The m th term becomes

$$k_{n-m} = \frac{1}{m!} \frac{d^m}{ds^m}[(s + p)^n F(s)] \big|_{s=-p}$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+a)^n}\right] = \frac{t^{n-1}e^{-at}}{(n-1)!}u(t)$$

$$f(t) = \left(k_1 e^{-pt} + k_2 t e^{-pt} + \frac{k_3}{2!} t^2 e^{-pt} \right. \\ \left. + \cdots + \frac{k_n}{(n-1)!} t^{n-1} e^{-pt} \right) u(t) + f_1(t)$$

Numerical - 7

Calculate $v(t)$ given that

$$V(s) = \frac{10s^2 + 4}{s(s + 1)(s + 2)^2}$$

Solution:

$$\begin{aligned} V(s) &= \frac{10s^2 + 4}{s(s + 1)(s + 2)^2} \\ &= \frac{A}{s} + \frac{B}{s + 1} + \frac{C}{(s + 2)^2} + \frac{D}{s + 2} \end{aligned}$$

■ **METHOD 1** Residue method:

$$A = sV(s) \Big|_{s=0} = \frac{10s^2 + 4}{(s + 1)(s + 2)^2} \Big|_{s=0} = \frac{4}{(1)(2)^2} = 1$$

$$B = (s + 1)V(s) \Big|_{s=-1} = \frac{10s^2 + 4}{s(s + 2)^2} \Big|_{s=-1} = \frac{14}{(-1)(1)^2} = -14$$

$$C = (s + 2)^2 V(s) \Big|_{s=-2} = \frac{10s^2 + 4}{s(s + 1)} \Big|_{s=-2} = \frac{44}{(-2)(-1)} = 22$$

$$\begin{aligned} D &= \frac{d}{ds}[(s + 2)^2 V(s)] \Big|_{s=-2} = \frac{d}{ds} \left(\frac{10s^2 + 4}{s^2 + s} \right) \Big|_{s=-2} \\ &= \frac{(s^2 + s)(20s) - (10s^2 + 4)(2s + 1)}{(s^2 + s)^2} \Big|_{s=-2} = \frac{52}{4} = 13 \end{aligned}$$

■ **METHOD 2** Algebraic method:

$$10s^2 + 4 = A(s + 1)(s + 2)^2 + Bs(s + 2)^2 \\ + Cs(s + 1) + Ds(s + 1)(s + 2)$$

$$10s^2 + 4 = A(s^3 + 5s^2 + 8s + 4) + B(s^3 + 4s^2 + 4s) \\ + C(s^2 + s) + D(s^3 + 3s^2 + 2s)$$

Equating coefficients,

$$\text{Constant: } 4 = 4A \quad \Rightarrow \quad A = 1$$

$$s: \quad 0 = 8A + 4B + C + 2D \quad \Rightarrow \quad 4B + C + 2D = -8$$

$$s^2: \quad 10 = 5A + 4B + C + 3D \quad \Rightarrow \quad 4B + C + 3D = 5$$

$$s^3: \quad 0 = A + B + D \quad \Rightarrow \quad B + D = -1$$

Solving these simultaneous equations gives $A = 1$, $B = -14$, $C = 22$, $D = 13$, so that

$$V(s) = \frac{1}{s} - \frac{14}{s+1} + \frac{13}{s+2} + \frac{22}{(s+2)^2}$$

Taking the inverse transform of each term, we get

$$v(t) = (1 - 14e^{-t} + 13e^{-2t} + 22te^{-2t})u(t)$$

Complex poles

- A pair of complex poles is simple if it is not repeated; it is a double or multiple pole if repeated.
- Simple complex poles may be handled the same way as simple real poles, but because complex algebra is involved the result is always cumbersome.
- An easier approach is a method known as completing the square.
- The idea is to express each complex pole pair in $D(s)$ as a complete square such as $(s + \alpha)^2 + \beta^2$

$$F(s) = \frac{A_1 s + A_2}{s^2 + as + b} + F_1(s)$$

where $F_1(s)$ is the remaining part of $F(s)$ that does not have this pair of complex poles. If we complete the square by letting

$$s^2 + as + b = s^2 + 2\alpha s + \alpha^2 + \beta^2 = (s + \alpha)^2 + \beta^2$$

$$A_1 s + A_2 = A_1(s + \alpha) + B_1 \beta$$

$$F(s) = \frac{A_1(s + \alpha)}{(s + \alpha)^2 + \beta^2} + \frac{B_1 \beta}{(s + \alpha)^2 + \beta^2} + F_1(s)$$

$$f(t) = (A_1 e^{-\alpha t} \cos \beta t + B_1 e^{-\alpha t} \sin \beta t) u(t) + f_1(t)$$

Numerical - 8

Find the inverse transform of the frequency-domain function

$$H(s) = \frac{20}{(s + 3)(s^2 + 8s + 25)}$$

Solution:

$H(s)$ has a pair of complex poles at $s^2 + 8s + 25 = 0$

$$s = -4 \pm j3.$$

$$H(s) = \frac{20}{(s + 3)(s^2 + 8s + 25)} = \frac{A}{s + 3} + \frac{Bs + C}{(s^2 + 8s + 25)}$$

Algebraic method:

$$\begin{aligned}20 &= A(s^2 + 8s + 25) + (Bs + C)(s + 3) \\&= A(s^2 + 8s + 25) + B(s^2 + 3s) + C(s + 3)\end{aligned}$$

Equating coefficients gives

$$s^2: \quad 0 = A + B \quad \Rightarrow \quad A = -B$$

$$s: \quad 0 = 8A + 3B + C = 5A + C \quad \Rightarrow \quad C = -5A$$

$$\text{Constant: } 20 = 25A + 3C = 25A - 15A \quad \Rightarrow \quad A = 2$$

That is, $B = -2$, $C = -10$. Thus,

$$\begin{aligned}H(s) &= \frac{2}{s + 3} - \frac{2s + 10}{(s^2 + 8s + 25)} = \frac{2}{s + 3} - \frac{2(s + 4) + 2}{(s + 4)^2 + 9} \\&= \frac{2}{s + 3} - \frac{2(s + 4)}{(s + 4)^2 + 9} - \frac{2}{3} \frac{3}{(s + 4)^2 + 9}\end{aligned}$$

Taking the inverse of each term, we obtain

$$h(t) = \left(2e^{-3t} - 2e^{-4t} \cos 3t - \frac{2}{3}e^{-4t} \sin 3t \right) u(t)$$